Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 6

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Count the Number of Characters in a Text File
Write a program to read a text file and count the total number of characters present in it, excluding newline characters.
Q2. Search for a Student in a Dictionary
Write a program to store student roll numbers and names in a dictionary. Read a roll number from the user and display the corresponding student name if it exists.
Q3. Display Employees with Salaries Above a Given Amount from a CSV File
Write a program to read employee names and salaries from week_6_set2_q3.csv and display the employees whose salaries are greater than a user-entered amount.
Q4. Count the Occurrences of Each Character in a Text File
Write a program to read a text file and use a dictionary to count the occurrences of each character, excluding spaces and newline characters.
Q5. Store and Display Product Prices Using a Dictionary
Write a program to store product names and their prices in a dictionary. Allow the user to enter a product name and display its price if it exists in the dictionary.
Q6. Find the Average Marks of Each Student from a CSV File
Write a program to read student names and marks from week_6_set2_q6.csv and use a dictionary to calculate and display the average marks of each student.
Q7. Remove Blank Lines from a Text File
Write a program to read a text file and write all non-empty lines into another text file, removing blank lines.
Q8. Store Student Names and Grades in a CSV File
Write a Python program that asks the user to enter the names and grades of five students. Store the data in a list of lists and write the results into a CSV file named week_6_set2_q8.csv.
Solutions
1. Count the Number of Characters in a Text File

```
file = open("story.txt", "w")
file.write("Python is easy\n")
file.write("Python is useful")
file.close()

file_name = input("Enter file name: ")

file = open(file_name, "r")
```

```
text = file.read()
file.close()

count = 0

for ch in text:
    if ch != "\n":
        count += 1

print("Total number of characters:", count)
```

Output:

```
Enter file name: story.txt
Total number of characters: 30
```

2. Search for a Student in a Dictionary

```
students = {
    101: "Aarav",
    102: "Diya",
    103: "Kabir",
    104: "Meera",
    105: "Rohan"
}

roll_no = int(input("Enter roll number: "))

if roll_no in students:
    print("Student name:", students[roll_no])
else:
    print("Student not found")
```

Output 1:

```
Enter roll number: 103
Student name: Kabir
```

Output 2:

```
Enter roll number: 110
Student not found
```

3. Display Employees with Salaries Above a Given Amount from a CSV File

```
import csv

file = open("week_6_set2_q3.csv", "r")
reader = csv.DictReader(file)

amount = int(input("Enter salary amount: "))
```

```
print("Employees earning above", amount, ":")

for row in reader:
    name = row["Employee Name"]
    salary = int(row["Salary"])

    if salary > amount:
        print(name, ":", salary)

file.close()
```

Output:

```
Enter salary amount: 45000
Employees earning above 45000 :
Rahul : 55000
Ishita : 48000
Vikram : 62000
```

4. Count the Occurrences of Each Character in a Text File

```
file = open("characters.txt", "w")
file.write("hello python")
file.close()

file_name = input("Enter file name: ")

file = open(file_name, "r")
text = file.read()
file.close()

frequency = {}

for ch in text:
    if ch != " " and ch != "\n":
        if ch in frequency:
            frequency[ch] += 1
        else:
            frequency[ch] = 1

print("Character frequencies:")

for ch in frequency:
    print(ch, ":", frequency[ch])
```

Output:

```
Enter file name: characters.txt
```

```
Character frequencies:
h : 2
e : 1
l : 2
o : 2
p : 1
y : 1
t : 1
n : 1
```

5. Store and Display Product Prices Using a Dictionary

```
products = {
    "Notebook": 45,
    "Pen": 10,
    "Backpack": 850,
    "Marker": 25,
    "Eraser": 5
}

print("Available products:")

for name in products:
    print(name, ":", products[name])

product_name = input("Enter product name: ")

if product_name in products:
    print("Price of", product_name, ":", products[product_name])
else:
    print("Product not found")
```

Output 1:

```
Available products:
Notebook : 45
Pen : 10
Backpack : 850
Marker : 25
Eraser : 5
Enter product name: Pen
Price of Pen : 10
```

Output 2:

```
Available products:
Notebook : 45
Pen : 10
```

```
Backpack : 850
Marker : 25
Eraser : 5
Enter product name: Pencil
Product not found
```

6. Find the Average Marks of Each Student from a CSV File

```
import csv

file = open("week_6_set2_q6.csv", "r")
reader = csv.DictReader(file)

students = {}

for row in reader:
    name = row["Student Name"]
    marks = int(row["Marks"])

    if name in students:
        students[name].append(marks)
    else:
        students[name] = [marks]

file.close()

print("Average marks of each student:")

for name in students:
    total = 0

    for mark in students[name]:
        total += mark

    average = total / len(students[name])

    print(name, ":", round(average, 2))
```

Output:

```
Average marks of each student:
Aarav : 83.0
Diya : 88.0
Kabir : 88.0
Meera : 79.0
```

7. Remove Blank Lines from a Text File

```
file = open("input.txt", "w")
file.write("Python is easy\n")
file.write("\n")
file.write("Python uses indentation\n")
file.write("   \n")
file.write("Python supports dictionaries\n")
file.close()

input_name = input("Enter input file name: ")
output_name = input("Enter output file name: ")

source = open(input_name, "r")
destination = open(output_name, "w")

for line in source:
    if line.strip() != "":
destination.write(line)

source.close()
destination.close()

print("File after removing blank lines:")

file = open(output_name, "r")
print(file.read(), end="")
file.close()
```

Output:

```
Enter input file name: input.txt
Enter output file name: output.txt
File after removing blank lines:
Python is easy
Python uses indentation
Python supports dictionaries
```

8. Store Student Names and Grades in a CSV File

```
import csv

students = []

for i in range(5):
    name = input("Enter student name: ")
    grade = input("Enter grade (A/B/C): ")
students.append([name, grade])
```

```
file = open("week_6_set2_q8.csv", "w", newline="")
writer = csv.writer(file)

writer.writerow(["Student Name", "Grade"])
writer.writerows(students)

file.close()

print("Data written to week_6_set2_q8.csv")

file = open("week_6_set2_q8.csv", "r")
reader = csv.reader(file)


for row in reader:
    print(row[0], row[1])

file.close()
```

Output:

```
Enter student name: Aarav
Enter grade (A/B/C): A
Enter student name: Diya
Enter grade (A/B/C): B
Enter student name: Kabir
Enter grade (A/B/C): A
Enter student name: Meera
Enter grade (A/B/C): C
Enter student name: Rohan
Enter grade (A/B/C): B
Data written to week_6_set2_q8.csv
Student Name, Grade
Student Name Grade
Aarav A
Diya B
Kabir A
Meera C
Rohan B
```

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